ASSIGNMENT 9

ADVANCED SQL QUERIES USING JOINS AND SUBQUERIES

Project: Library Management System | Database: LibraryManagementSystem

Student Name: _____ **Class:** _____

1. Introduction

Advanced SQL queries retrieve related information from multiple tables and generate reports. This assignment uses INNER JOIN, LEFT JOIN and subqueries to analyse loans, members, books, staff, reservations and fines.

Sample-data note: The outputs in this report are illustrative expected results based on a consistent example dataset: 5 book records (total copies 5, 4, 3, 6 and 5; category counts 1, 3 and 1), 4 members, 2 staff members, 6 loans (2 per first two members, 1 per next two members), 3 reservations and 2 fines (■20.00 and ■15.00). Run the queries against matching inserted data to reproduce these outputs. Dates and names are example data.

2. Types of Joins and Subqueries

INNER JOIN: Returns rows where matching values exist in both joined tables.
LEFT JOIN: Returns every row from the left table and matching rows from the right table. If no match exists, right-table columns contain NULL.
Subquery: A query nested inside another query. Its result is used by the outer query.

3. Loan Report – Books Borrowed by Members

Query 1: Display loan details with book titles and member names.

```
SELECT
    L.LoanID,
    B.Title,
    M.FullName,
    L.IssueDate,
    L.DueDate,
    L.Status
FROM Loans L
INNER JOIN Books B ON L.BookID = B.BookID
INNER JOIN Members M ON L.MemberID = M.MemberID
ORDER BY L.LoanID;
```

LoanID	Title	FullName	IssueDate	DueDate	Status
1001	Database Management Systems	Asha Sharma	2026-10-01	2026-10-15	Returned
1002	Python Programming	Rahul Verma	2026-10-05	2026-10-19	Issued
1003	Artificial Intelligence	Asha Sharma	2026-09-20	2026-10-04	Returned
1004	Database Management Systems	Priya Nair	2026-10-10	2026-10-24	Issued
1005	Java Programming	Karan Rao	2026-09-25	2026-10-09	Returned
1006	Computer Networks	Rahul Verma	2026-10-12	2026-10-26	Issued

Explanation: INNER JOIN combines Loans with Books and Members using their foreign-key relationships. The report identifies the book and member for every loan.

4. Fine Report – Member and Book Details

Query 2: Display fine records with the member and book associated with each loan.

```
SELECT
    F.FineID,
    M.FullName,
    B.Title,
    L.DueDate,
    F.Amount,
    F.PaidStatus
FROM Fines F
INNER JOIN Loans L ON F.LoanID = L.LoanID
INNER JOIN Members M ON L.MemberID = M.MemberID
INNER JOIN Books B ON L.BookID = B.BookID
ORDER BY F.FineID;
```

FineID	FullName	Title	DueDate	Amount	PaidStatus
5001	Asha Sharma	Artificial Intelligence	2026-10-04	20.00	Paid
5002	Karan Rao	Java Programming	2026-10-09	15.00	Unpaid

Explanation: This query joins Fines to Loans, Members and Books. It shows who incurred the fine and which book it relates to.

5. Member Loan Summary – LEFT JOIN

Query 3: Display every member and the number of loan records for that member.

```
SELECT
    M.MemberID,
    M.FullName,
    COUNT(L.LoanID) AS TotalLoans
FROM Members M
LEFT JOIN Loans L ON M.MemberID = L.MemberID
GROUP BY M.MemberID, M.FullName
ORDER BY M.MemberID;
```

MemberID	FullName	TotalLoans
201	Asha Sharma	2
202	Rahul Verma	2
203	Priya Nair	1
204	Karan Rao	1

Explanation: LEFT JOIN ensures that members are included even if they have no loan records. COUNT(L.LoanID) returns zero for a member with no matching loan.

6. Subquery – Books Above Average Stock

Query 4: Display books whose total copies are greater than the average.

```
SELECT BookID, Title, TotalCopies
FROM Books
WHERE TotalCopies > (
    SELECT AVG(TotalCopies)
    FROM Books
)
ORDER BY BookID;
```

BookID	Title	TotalCopies
101	Database Management Systems	5
104	Computer Networks	6
105	Java Programming	5

Explanation: The inner query calculates the average number of total copies. The outer query returns books with a value greater than that average (4.6 in the example data).

7. Subquery with GROUP BY and HAVING – Frequent Borrowers

Query 5: Display members whose loan count is greater than the average loan count per borrowing member.

```
SELECT
    M.MemberID,
    M.FullName,
    COUNT(L.LoanID) AS TotalLoans
FROM Members M
INNER JOIN Loans L ON M.MemberID = L.MemberID
GROUP BY M.MemberID, M.FullName
HAVING COUNT(L.LoanID) > (
    SELECT AVG(LoanCount)
    FROM (
        SELECT COUNT(*) AS LoanCount
        FROM Loans
        GROUP BY MemberID
    ) AS MemberLoanStats
)
ORDER BY M.MemberID;
```

MemberID	FullName	TotalLoans
201	Asha Sharma	2
202	Rahul Verma	2

Explanation: The nested query counts loans for each member and calculates the average of those counts. The outer query displays members whose count is above that average. Members with no loans are not included in this average.

8. Reservation Report

Query 6: Display reservation details with member names and book titles.

```
SELECT
    R.ReservationID,
    M.FullName,
    B.Title,
    R.ReservationDate,
    R.Status
FROM Reservations R
INNER JOIN Members M ON R.MemberID = M.MemberID
INNER JOIN Books B ON R.BookID = B.BookID
ORDER BY R.ReservationID;
```

ReservationID	FullName	Title	ReservationDate	Status
6001	Priya Nair	Artificial Intelligence	2026-10-20	Pending
6002	Karan Rao	Database Management Systems	2026-10-21	Confirmed
6003	Asha Sharma	Python Programming	2026-10-22	Pending

Explanation: The query combines Reservations with Members and Books to identify the member and book associated with each reservation.

9. Staff Loan Report

Query 7: Display loan transactions handled by each staff member.

```
SELECT
    L.LoanID,
    S.FullName AS StaffName,
    M.FullName AS MemberName,
    B.Title,
    L.Status
FROM Loans L
INNER JOIN Staff S ON L.StaffID = S.StaffID
INNER JOIN Members M ON L.MemberID = M.MemberID
INNER JOIN Books B ON L.BookID = B.BookID
ORDER BY L.LoanID;
```

LoanID	StaffName	MemberName	Title	Status
1001	Meera	Asha Sharma	Database Management Systems	Returned
1002	Meera	Rahul Verma	Python Programming	Issued
1003	Arjun	Asha Sharma	Artificial Intelligence	Returned
1004	Arjun	Priya Nair	Database Management Systems	Issued
1005	Meera	Karan Rao	Java Programming	Returned
1006	Arjun	Rahul Verma	Computer Networks	Issued

Explanation: This report joins four tables to show which staff member handled each loan, which member borrowed the book and the current loan status.

10. Conclusion

Joins and subqueries help retrieve useful information from related database tables. The reports in this assignment show loan details, fine records, member borrowing activity, reservations and staff transactions. These queries demonstrate how SQL can support reporting and decision-making in the Library Management System.